

Acoustic design evaluation in educational buildings using artificial intelligence

Mohammad Tabatabaei Manesh^a, Arman Nikkhah Dehnavi^b, Mohammad Tahsildoost^b,
Pantea Alambeigi^{c,*}

^a College of Built Environments, University of Washington, Seattle, USA

^b School of Architecture and Urban Planning, Shadid Beheshti University, Tehran, Iran

^c School of Design and Architecture, Swinburne University, Melbourne, Australia

ARTICLE INFO

Keywords:

Building acoustics
Catboost
Pix2pix
Educational building
Speech intelligibility

ABSTRACT

Speech intelligibility is a critical aspect of building science, particularly in educational buildings where poor sound quality may have a detrimental impact on students' learning and teachers' health. However, considering the numerous building regulations proposing varying definitions and ranges of acoustic comfort, calculating the necessary acoustic indicators can be challenging for designers. Speech intelligibility is a crucial component of indoor acoustics and acoustic comfort and can be calculated using formulas, simulation software, and data-based web tools. While formulas are fast, they lack details; acoustic simulation software is highly accurate but time-consuming and expensive. Data-based web tools, including machine learning algorithms, offer both speed and accuracy and are widely accessible. In this study, we present a system utilizing machine learning techniques to predict acoustic indicators, numeric and heatmap, in an educational building. The Pachyderm plugin in the Grasshopper was utilized to conduct extensive simulations in a single educational space, focusing on acoustic indicators in six different frequencies and general modes. Using Catboost and the pix2pix algorithm, the prediction models provide numerical and image indices on the developed dataset. Also, SHAP values were employed to interpret the Catboost model, analyzing the significance of each feature. The results showed remarkable accuracy, (i.e., between 89 % and 99 %) in the numerical portion, and PSNR index ranging from 0.817 to 0.970, and an SSIM index ranging from 15.56 to 31.57 in the image section. By utilizing data-driven methods, the system provides an efficient and accurate approach to calculating acoustic indicators, helping to ensure optimal acoustic environment in educational buildings.

1. Introduction

In the evolving landscape of educational architecture, the significance of acoustic comfort has increasingly come to the fore. Despite its acknowledged significance as a crucial element [1], the concept lacks a universally agreed-upon definition and thorough exploration in literature. Although recognized as pivotal, it often remains a lower priority in the design and renovation of buildings [2]. Noise was found to be the most annoying factor in educational buildings [3], which as a significant stressor, stimulates both the sympathetic nervous system and the endocrine system [4]. Conversely, suitable acoustic conditions in indoor settings can significantly contribute to enhancing the well-being of individuals [5]. Previous studies showed that the interplay of sound and

space within educational environments not only influences the learning experience and performance but also profoundly affects the well-being of both teachers and students [6,7]. The most frequently reported issues include high noise levels in classrooms and reverberant environments, which make it difficult for students to hear and understand conversations and discussions, thus negatively impacting their learning experience. For teachers, these conditions can result in health problems such as increased stress levels, vocal fatigue due to raised vocal effort, and a greater propensity to consider leaving the profession. Vocal effort is associated with both visual and auditory perception of the space [8] and has a direct relationship with room's acoustics [9], in particular room's reverberation time (RT) [10].

When optimizing indoor acoustics, it is crucial to accommodate the

* Corresponding author.

E-mail addresses: mhtaba@uw.edu (M. Tabatabaei Manesh), a.nikkhahdehnavi@mail.sbu.ac.ir (A. Nikkhah Dehnavi), m_tahsildoost@sbu.ac.ir (M. Tahsildoost), palambeigi@swin.edu.au (P. Alambeigi).

<https://doi.org/10.1016/j.buildenv.2024.111695>

Received 28 January 2024; Received in revised form 2 May 2024; Accepted 29 May 2024

Available online 3 June 2024

0360-1323/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

diverse preferences and needs of students and teachers regarding Indoor Environmental Quality (IEQ) factors [11], with a particular focus on acoustical preferences [12]. Appropriate acoustic design is vital for fostering an inclusive environment for international students and those with learning or hearing impairments [13]. Speech intelligibility within classrooms significantly affects the quality of the student experience [14]. Therefore, classroom designs should incorporate acoustical parameters, such as the Clarity index and the Speech Transmission Index, to maintain adequate speech intelligibility levels [15].

Each country's approach reflects its specific concerns and priorities regarding the acoustic environment in educational settings such as Australia [16], Japan [17], UK [18] and USA [19]. Despite the differences in standards, the overarching goal remains consistent: to create conducive learning environments by optimizing acoustical quality to support effective teaching and learning. The diversity in approaches underscores the complexity of the issue and the importance of tailored solutions to meet the unique needs of educational facilities in different regions.

Various room acoustic parameters are identical in the mentioned standards, as shown in Table 1. By analyzing parameters such as Reverberation Times (T_{30}), Early Decay Time (EDT), percentage articulation loss of consonants (%ALC), Definition (D_{50}), Speech Transmission Index (STI), and Clarity (C_{80}) the acoustic properties of rooms can be determined. While there are various indicators available, T_{30} , EDT, C_{80} , D_{50} , and STI are among the most used in research in educational building and are calculated by most acoustic tools [20].

Analyzing acoustic conditions and developing predictive models are essential for designing buildings with optimal auditory environments [22]. Traditionally, evaluating acoustic design in buildings has been a time-consuming and resource-intensive process, relying on physical measurements and empirical calculations [23]. Moreover, achieving this was not feasible for architects without seeking guidance from acoustic engineers, thus impeding the incorporation of acoustic design during the preliminary stages. The emergence of advanced computational methods has provided architects with an opportunity to delve into acoustic design and revolutionize this field.

Acoustic simulation techniques, employing algorithms like

Geometrical Acoustics (GA) and numerical solutions such as the Finite Difference Method (FDM), Finite Element Method (FEM), and Boundary Element Method (BEM), are crucial for predicting and optimizing system acoustics [24]. Each technique comes with its own set of strengths and drawbacks, and the selection of a particular method is contingent on the unique problem at hand. FEA is suitable for simulating complex geometries and boundary conditions [24], while BEM is more efficient for problems with complex boundary conditions [25]. Ray tracing is useful for predicting the acoustic behavior of indoor spaces [26]. The Image-Source Method (ISM) is another GA method and crucial for indoor sound spread prediction, conceptualizing reflected sound rays as emanating from "image sources" formed by planar walls [27].

These methods form the foundation of popular commercial room acoustic simulation software such as ODEON, CATT-Acoustics, and Raven. The choice of numerical methods may vary based on the frequency range, geometry, and material properties of the problem. Table 2 presents various common acoustic tools, detailing their input/output parameters, along with their respective advantages and disadvantages.

Data-driven approaches have gained considerable traction and achieved noteworthy accomplishments due to the increasing availability of data [28]. In the realm of acoustics, advancements in machine learning are progressing rapidly, demonstrating impressive results and promising potential for future developments. In architectural acoustics, Nannariello & Fricke [29] applied neural networks to predict reverberation time in 71 spaces, such as concert halls, auditoriums, and cultural centers. By comparing the reverberation time predicted by the neural network with that calculated using the Sabine formula and ODEON 2.6D, researchers were able to assess the predictive accuracy of the trained model. Falcon Perez [30] conducted a study on the acoustic indicators of a single space and developed a machine-learning predictive model that considered various indoor characteristics, such as the size and placement of furniture. Abarghoie et al. [31] presented a working prototype for a new method of machine learning to approximate a series of typical room acoustic parameters using only geometric data as input characteristics. A dataset as the input of seven ML models was generated based on fully connected Deep Neural Networks (DNN). The average error of ML models was between 1 % and 3 %. Yeh & Tsay [32] employed machine learning to develop a predictive model for acoustic indicators, designed as a simple evaluation tool for the architectural design and interior decoration of multi-functional activity centers. They created 800 spaces using parametric design and utilized Odeon to gather acoustic indicators. By applying machine learning techniques, they formulated a convenient, quick, and accurate prediction model. This method allowed for the collection of various acoustic metrics for the spaces, removing the need for 3D modeling or simulation software.

Educational buildings, which serve varied functions, are significantly influenced by elements such as wall material, room dimensions, the number of furnishings, and ceiling material, as found by Abarghoie et al. [31]. The accuracy of these elements is crucial, as inexperienced design estimates can lead to costly and time-consuming rework. Traditional methods, such as quick formulas, often miss crucial details, and while more accurate analyses can be obtained through acoustic simulation software like Odeon or Ease, these processes are time-intensive and seldom used in practical scenarios. Effectively improving indoor acoustics at the early design stages involves rapidly comparing various options with reasonable accuracy to optimize outcomes.

This research enhances architectural design in educational environments by integrating theoretical and empirical analyses with user-friendly, machine learning tools to streamline acoustic evaluations. It aims to simplify the estimation of crucial indoor acoustic indexes without requiring complex acoustic calculations. Utilizing techniques such as statistical data extraction and data-driven methods, including machine learning and deep learning, architects can design using standard acoustic indexes, focusing on elements like dimensions, materials, or source positions, without needing extensive acoustic knowledge. This study addresses significant gaps in previous research by expanding the

Table 1
Common indoor room acoustic parameters in the building standards.

Index	Speech	Music	Description
Reverberation Time (T_{30})	✓	✓	The time required for the sound pressure level to fall 60 dB [16].
Early Decay Time (EDT)	✓ (1s)	✓ (1–2 s)	Evaluated from a linear regression of the first 10 dB of decay of the backward integrated squared impulse response curve [17].
Clarity (C_{80})	✓ (>3 dB)	✓ (–4 to 2 dB)	The ratio of early received sound energy (measured within 0–80 ms after the arrival of direct sound) to the total received sound energy [18].
Definition (D_{50})	✓ (>50 %)	✓ (<50 %)	The ratio of the early received sound energy (0–50 ms after direct sound arrival) to the total received energy [19].
Speech Transmission Index (STI)	✓ (>0.5)	–	Transmission of a speech signal between a speaker and a listener [20].
Center Time (T_s)	✓ (60–80 ms)	✓ (70–150 ms)	Corresponds to the center of gravity of the squared impulse response [17].
Articulation Index (AI)	✓	–	A psychoacoustic metric that indicates how much interference noise can interfere with human speech [21].
Privacy Index (PI)	✓	–	Based on a mathematical relation with AI, developed to rate communication rather than privacy.

Table 2

A comparative overview of common building acoustic software tools: Features, capabilities, and performance.

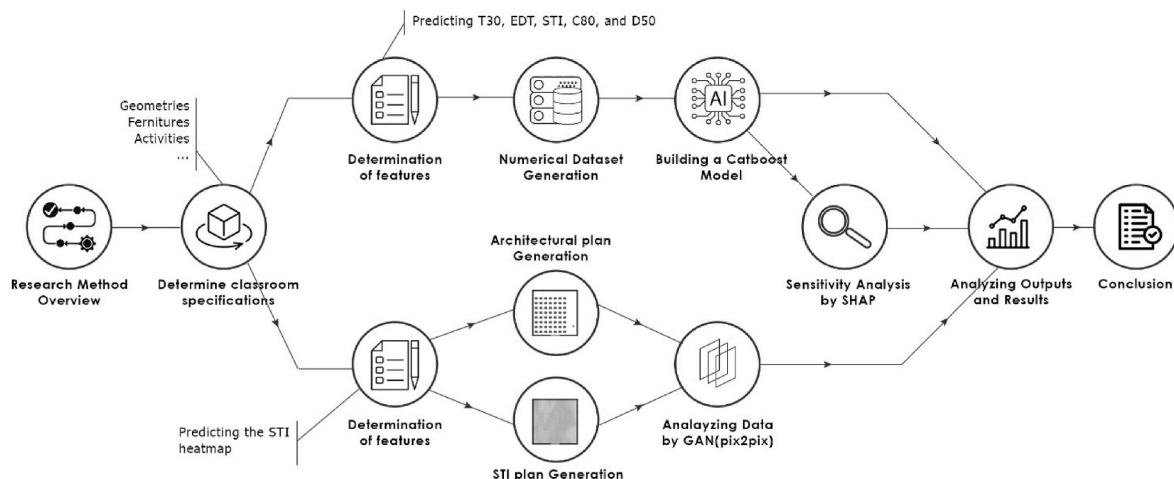
Software	Parameter				Indices	Features						
	Indoor acoustic	Noise control	transmission	Sound propagation		Material library	Complexity	3D model	Accuracy	Calculation time	Cost	Other
ODEON	✓	✓	✓	✓	SPL, SPL (A), DL2, T ₃₀ , EDT, STI, Ts, G, C ₈₀ , D ₅₀ , LF ₈₀ , ST, LG ₈₀ LwA	✓	advanced	✓	high	medium	✓	–
EASE	✓	×	×	×	RT, EDT, C ₈₀ , C ₅₀ , STI, ALCons, SPL, D/R, LF, EDC	✓	advanced	✓	high	high	✓	Hardware Requirements
Pachyderm	✓	×	×	×	SPL, C _t , C, D ₅₀ , EDT, T, STI	×	medium	✓	medium	high	free	Open-source, parametric simulation
INSUL	✓	×	✓	×	STC, R _w , IIC, Ln _{TW}	✓	advanced	✓	high	medium	✓	Limited scope, Data requirements
CATT-Acoustic	✓	✓	×	✓	RT, EDT, C, STI, SPL, DRR, ETC	✓	medium	×	medium	high	✓	–
DIRAC	✓	×	×	×	RT, EDT, STI, SPL	✓	advanced	×	medium	high	✓	sound system optimization
Troldtekt, Jocavi, Dampa	✓	×	×	×	T, SPL	×	easy	×	low	low	free	Quick
EXNO	×	×	✓ (Windows)	×	R _w	✓	easy	✓	medium	medium	free	Limited scope and indicators

scope of features beyond mere geometry, thereby providing a more comprehensive analysis and a broader range of design options. Additionally, it introduces a novel method for generating acoustic heatmaps from architectural plans, offering designers a holistic view of the acoustic environment, enhancing their ability to make informed decisions during the initial design stages. This approach not only facilitates quicker, more accurate design processes but also contributes to creating sustainable, human-centric educational spaces.

2. Materials and methods

The current study employs a methodology that involves multiple stages. Two types of datasets [33] were generated: numerical indoor acoustic indices and STI heat maps. In both types, Grasshopper and

Pachyderm plugins were used to generate data. These plans and their acoustic simulation results were further used as a dataset which was applied for training the machine learning algorithm. The plans were simulated in the Pachyderm, and the obtained results were recorded and analyzed through two main procedures. First, a Pix2Pix algorithm underwent training using a dataset consisting of 336 pairs of images. Each pair consisted of a plan and its corresponding heat map, both derived from the simulations. Each pair of images was sized at 256px × 512px. At this stage, only the heatmaps in the classroom were recognized, and the output did not include any details pertaining to other numerical indices related to indoor acoustics. In the second phase of the study, a dataset consisting of numerical values was compiled. Using this dataset, a Catboost predictive model was developed to forecast the values of T₃₀, EDT, STI, C₈₀, and D₅₀. This model was instrumental in identifying and

**Fig. 1.** Research method overview.

quantifying the key factors that influence the acoustic quality within a classroom environment. Following the implementation of the second stage, the study utilized SHAP (Shapley Additive Explanations) values, a method for interpreting artificial intelligence models, to analyze the significance of each feature. This analysis focused on understanding both the positive and negative impacts of these features (*i.e.* increasing or decreasing Sound power) (Fig. 1).

The parametric simulation for the models took 30 days due to the time-consuming acoustic ray tracing calculations. The calculation method employed is a choice between ray tracing and image source analysis. For this simulation, a total of 10,000 rays are utilized to model the scenario. The simulation is set to conclude after 1000 cut-off times. In terms of environmental conditions, the air temperature is fixed at 20 °C, with a relative humidity of 50 %. Additionally, the static air pressure is maintained at 1000 hPa. The method of calculation is the combination of ray tracing and image source in the Pachyderm environment. It is assumed that the background noise is low and follows NC30. The values provided by Marsh [34] have been used to select the absorption coefficient for all materials at each frequency. The recommended values in Odeon software have been utilized for the scattering coefficient [35].

2.1. STI heatmaps

Given that the majority of educational spaces adopt a room layout with a rectangular plan, the geometry of the classroom in this study was similarly approached. The dimensions of the room were treated as variable along two directions, denoted as *x* and *y*.

Since the dataset consists of architectural plans, typically cut at a height of 1–1.5 m, variables such as height and roof material cannot be directly measured. Therefore, their values are assumed to be constant. Similarly, to streamline the simulation process and minimize computational time, the floor material has been treated as constant.

The receiver's position is located at 700 mm and at a height of 1150 mm. It forms a network of points that can present STI heatmaps (Fig. 2).

To predict STI heatmaps, it is necessary to assign custom color codes to variables that are not visibly changing in the plan, such as building materials or audio source power so that the machine learning algorithm can distinguish them from each other (Table 3).

2.2. Numerical indoor acoustic indices

To calculate other indoor acoustic indices in specific locations in the classroom and fix the defects of the image dataset, a numerical dataset was generated. In addition to other geometry features, the height of the room was also considered as a variable in two opposite directions, H1, and H2. The position of the speaker was considered to be four positions, which are placed linearly and in front of the listeners. Also, nine variable modes were defined for the position of the listeners in the entire class

Table 3

Variables colors in architectural layout plans.

Type	Information	Color	RGB code
Wall material	Plaster on brick or concrete wall		0,0,0
	Hollow concrete block without paint		184,0,125
	Concrete block with 200 ml paint		255,255,255
Window material	4 mm glass		13,222,20
	Double-glazed glass		30,67,130
Furniture material	Chair with fabric cover		77,23,212
	Wooden chair		214,188,101
Human	–		91,227,172
Source power	Normal		212,23,23
	Raised		148,189,121
	Loud		23,143,212
	Shouted		212,110,198

space (Fig. 3).

The power of the sound source, which is equal to the sound intensity in eight frequencies and in four modes of Normal, Raised, Loud, and Shouted, is defined as a variable [36]. The population of students in four modes is considered as a variable. The placement of people in the space is considered randomly in the entire space (Table 4).

2.3. Pix2pix training

Pix2Pix is a method that performs pixel-to-pixel translation using two input images. It comprises a generator and a discriminator. The discriminator assesses pairs of images, both generated and original, through multiple components like convolution layers and activation functions. The generator, employing a U-net structure with skip connections, aims to produce output resembling the actual data by transferring information between layers effectively. This setup ensures precise learning and representation of the relationship between different types of data, such as architectural plans and heat maps. The Pix2Pix

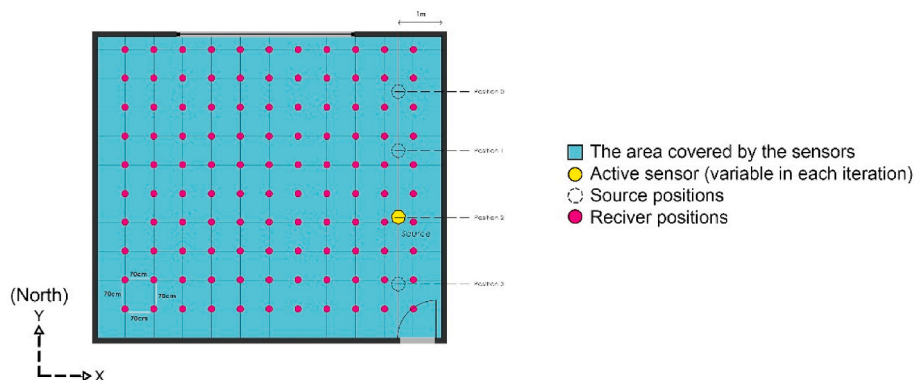


Fig. 2. Plan view of receivers and sources in STI heatmaps simulation.

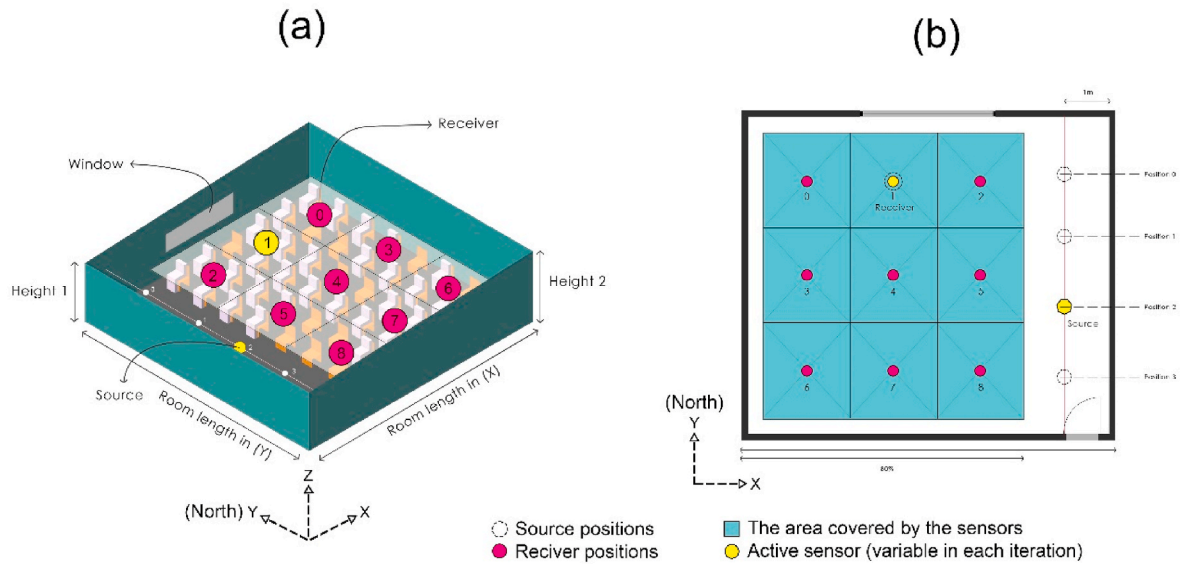


Fig. 3. 3D (a) and plan (b) view of receivers and sources in numerical indoor acoustic simulation.

Table 4
Features of defined spaces.

Variables	Domain	Step	Unit or modes	Number of modes
Room length in X direction	8–28	0.5	meters	40
Room length in Y direction	3–14	0.5	meters	22
Height 1 (h1)	2–4.5	0.5	meters	6
Height 2 (h2)	2–4.5	0.5	meters	6
WWR	20–80	20	%	4
Source power	–	–	1. Normal 2. Raised 3. Loud 4. Shouted	4
Receiver position	1–9	1	position	9
Source position	1–4	1	position	4
Population	25–100	25	%	4
Floor material	–	–	1. Stone and ceramics 2. Wooden flooring 3. 7 mm thick carpet installed on foam	3
Ceiling material	–	–	1. Painted concrete slab 2. 12 ml single layer gypsum board 3. Acoustic tile with mineral fibers 18 ml	3
Wall material	–	–	1. Plaster on brick or concrete wall 2. Hollow concrete block without paint 3. Concrete block with 200 ml paint	3
Window material	–	–	1. 4 mm glass 2. Double-glazed glass	2
Furniture material	–	–	1. Chair with fabric cover 2. Wooden chair	2

model comprehends these correlations, yielding meaningful and coherent outputs. The network's structure is illustrated in Fig. 4.

The encoder network in this setup is composed of seven convolutional blocks. Each block is structured with a convolutional layer, succeeded by a LeakyReLU activation function, and a batch normalization layer, with the exception of the first block. Conversely, the decoder network comprises seven transposing convolutional blocks, each followed by a convolutional layer, a batch normalization layer, and a ReLU

activation function. Implementing the Pix2Pix algorithm required pairing each architectural plan image with its corresponding simulation output, produced by Pachyderm. Utilizing the PILLOW library in Python, all sample plans and their respective heat maps were resized to dimensions of 256px by 256px. Subsequently, each plan was matched with its specific simulation output. The final dataset, as a result, consists of 336 paired images, each paired measuring 256px by 512px (Fig. 5). The data were randomly split into two sets: 90 % for the training and 10 % for the testing.

The entire training process was executed on a system consisting of 16 GB DDR3 RAM, a 2.5 MHz core i7 CPU, and the NVIDIA GeForce GTX 960 M graphics coprocessor.

To evaluate the accuracy of the predictions made by Pix2Pix, two metrics were used: the Structural Similarity Index Measure (SSIM) and the Peak-Signal-to-Noise Ratio (PSNR). SSIM is a highly regarded quality metric that measures the similarity between two images. Initially proposed by Wang et al., SSIM has demonstrated a strong correlation with the human visual system's perception [37].

The SSIM value ranges from 0 to 1, with a value of 1 indicating a perfect match in terms of reconstructing the image compared to the original one. Conversely, that greater PSNR values correspond to superior image quality. Conversely, a low PSNR value indicates significant numerical disparities between images [38].

2.4. Acoustic indices prediction through the catboost method and model explanation

Utilizing Pix2Pix, STI heat maps were generated for each space. The heatmaps do not give information about T_{30} , EDT, STI, C_{80} , and D_{50} values. The estimation of These indices could be considered a regression problem. Therefore, numerical data translations are applied in the current study to predict the Indoor acoustic indices in the plans as well. The T_{30} , EDT, C_{80} , and D_{50} values are calculated in six typical octave frequency bands (125, 250, 500, 1000, 2000, and 4000 Hz), while the STI index is computed in general, men's and women's mode. The data are obtained from acoustical simulations of a full 3D geometrical model of a classroom with simplified furniture and consideration of the absorption coefficient of the components. Therefore, the dataset consists of 980 rows, 14 columns as features, and 27 columns as the targets. In the current study, to elucidate the reasons and mechanisms by which specific features influence the output (i.e. STI), the Catboost method, combined with SHAP, was adopted.

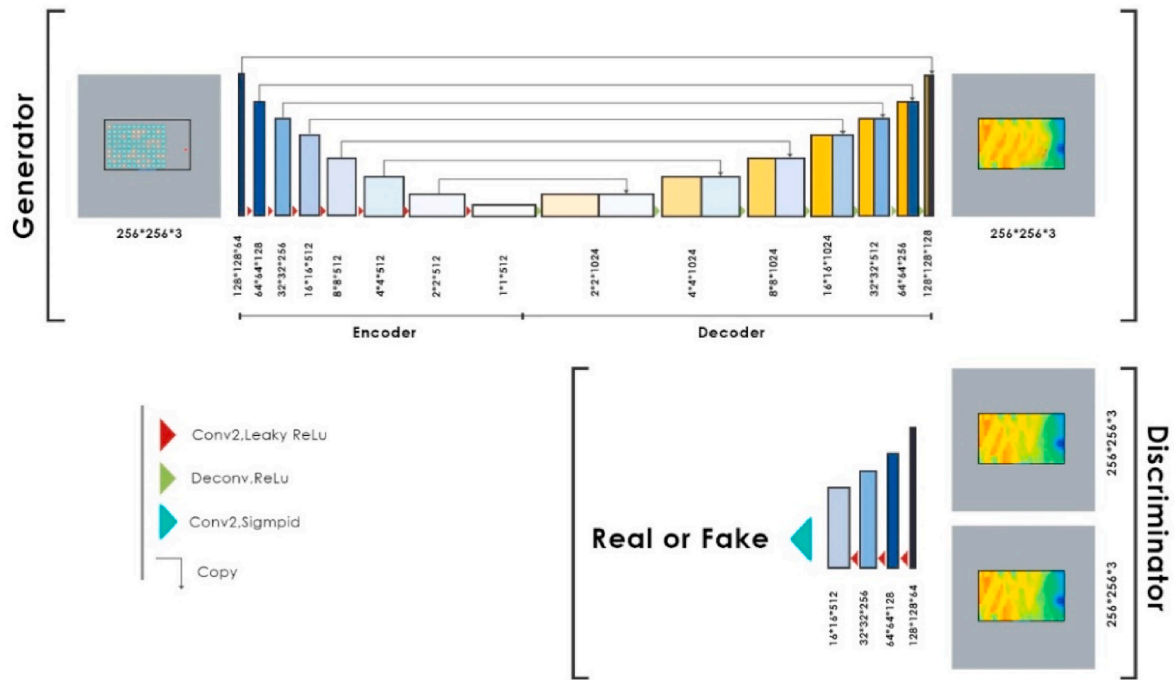


Fig. 4. The architecture of the Pix2Pix model.

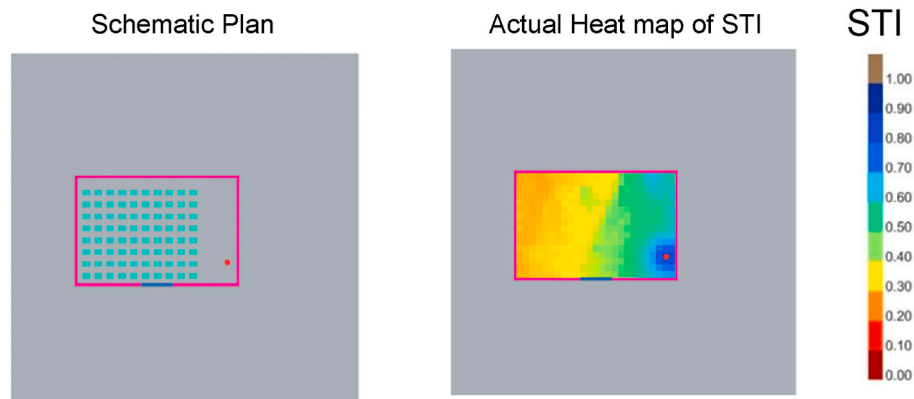


Fig. 5. A single data sample created for Pix2Pix training.

Catboost is a Gradient Boosting Decision Tree (GBDT) developed by Prokhorenkova [39] that supports categorical and numerical features as well as texts and embeddings. The data was split randomly into two groups: 85 % for training and 15 % for testing. The accuracy of the Catboost predictions in this study was assessed using three distinct error functions: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R2) [40].

The SHAP method, introduced by Lundberg and Lee is a technique for elucidating the influence of various features on the target variable [41]. Originating from the Shapley value in cooperative game theory, the SHAP approach evaluates an algorithm's behavior in two contrasting conditions: one where the target variable is absent and another where it is present. This technique helps identify the importance of each feature and sheds light on how each one contributes to the model's output [42], either by amplifying (positive effect) or diminishing (negative effect) it [41].

3. Results and discussion

The current study's findings are divided into three main sections:

firstly, the creation of a predictive model to produce heat maps for STI within an educational facility; secondly, the development of a model for predicting indoor acoustic indices at six standard frequencies; and thirdly, an analysis of the impact of architectural elements on these acoustic indices.

After the training phase, Catboost demonstrates remarkable efficiency by predicting Indoor Acoustic Indices in under a second. Similarly, Pix2Pix can swiftly generate heat maps for architectural layouts, taking less than 5 s to process every ten samples. This speed indicates that, on identical hardware, the traditional methods of modeling and simulation would consume significantly more time compared to the proposed method. Therefore, for designers and analysts, the methods introduced in this research offer a much faster alternative, enhancing efficiency in their work processes.

The PSNR and SSIM metrics for the generated 51 test Pix2Pix model are in a range of 15.56–31.57 and 0.81 to 0.97, respectively. The quality of the provided findings increases with larger values of PSNR and SSIM measures. Nonetheless, the permissible bounds significantly rely on the kind of produced data [43]. The error metrics indicate that the results have an acceptable level of accuracy given the size and constraints of the

dataset. The model can be trained with greater data variance as the dataset gets larger, which improves the accuracy of the output findings.

Table 5 shows some of the test samples generated. The model generated greater outcomes in lower-density occupancy combinations, where the plans of the classroom are elongated proportions. The most accurate results suggest that the model is sufficiently precise to be applied at the initial stages of schematic plan design for the sort of building under discussion. The employed metrics indicate that, despite a few remaining doubt points about the generated model’s reliability, it might be appropriately applied to any prepared dataset that makes use of the designers’ limitations.

The predictive model, trained using a dataset of regular geometrical shapes, has its accuracy directly influenced by this training data. Consequently, the proposed Pix2Pix model may struggle with predicting the heat-map pattern in models that incorporate layouts with more concave edges in their geometrical shapes. Altering the training dataset to include a wider variety of shapes can guide the model to better predict these more complex structures. Future research could delve into identifying which specific geometrical shapes are most challenging for the model and exploring strategies to enhance its accuracy in these

scenarios. This exploration is crucial for improving the model’s robustness and applicability to a broader range of architectural designs.

The MAE, MSE, and R2 factors for the Catboost predictive model’s accuracy of the T₃₀, EDT, C₈₀, D₅₀, and STI are presented in Table 6. The produced model is faster than conventional methods and can be used as a substitute for Pachyderm simulations.

The result of Catboost model testing showed a high accuracy in all indices and frequencies and could reach the similar range to the result of research by Abarghooie et al. [31]. However, the catboost model is much faster than methods like DNN and needs less computation.

SHAP can increase the interpretability of the “black box” of the ML models and was used to examine the primary factors influencing the Indoor acoustic indices in the simulated samples. The impact of each feature on the T₃₀, EDT, C₈₀, D₅₀, and STI is shown in Figs. 6–10.

The charts indicate that depending on the frequency, different architectural features exert varying degrees of influence on the model’s output. This kind of analysis is essential for understanding and interpreting the behavior of complex models in the context of architectural acoustics. The Shap values obtained reflect the relative importance of different factors across different frequency ranges. Notably, the higher

Table 5
Five samples of validation of the STI heatmap prediction.

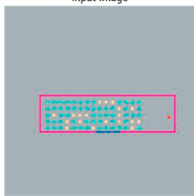
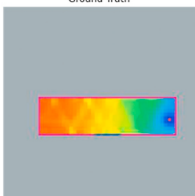
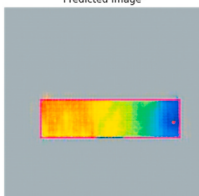
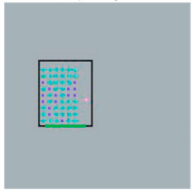
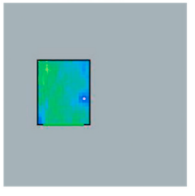
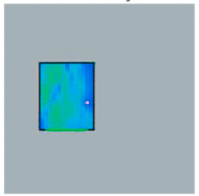
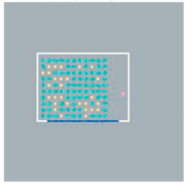
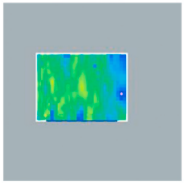
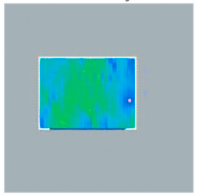
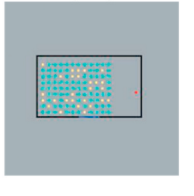
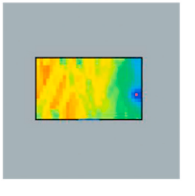
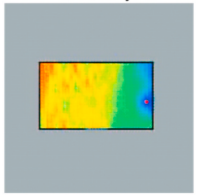
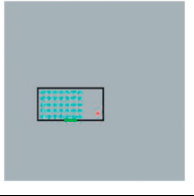
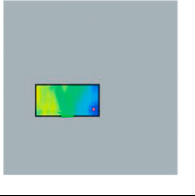
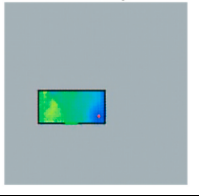
No.	Schematic Plan	Actual Heat map of STI	Predicted heat map of STI	PSNR	SSIM
1	Input Image 	Ground Truth 	Predicted Image 	20.76	0.884
2	Input Image 	Ground Truth 	Predicted Image 	20.00	0.90
3	Input Image 	Ground Truth 	Predicted Image 	15.56	0.81
4	Input Image 	Ground Truth 	Predicted Image 	19.85	0.84
5	Input Image 	Ground Truth 	Predicted Image 	24.38	0.94

Table 6Catboost model accuracy of T_{30} , C_{80} , D_{50} , EDT and STI.

Indices	Frequency/Mode	R2	MSE	MAE
T_{30}	125	0.976	0.006	0.055
	250	0.956	0.019	0.087
	500	0.912	0.037	0.113
	1000	0.894	0.036	0.115
	2000	0.928	0.019	0.091
C_{80}	4000	0.931	0.013	0.077
	125	0.984	0.322	0.365
	250	0.977	0.312	0.354
	500	0.917	0.371	0.441
	1000	0.911	0.345	0.454
D_{50}	2000	0.971	0.345	0.449
	4000	0.974	0.228	0.369
	125	0.987	4.293	1.549
	250	0.979	4.685	1.601
	500	0.920	7.851	2.180
EDT	1000	0.917	8.324	2.235
	2000	0.973	7.387	2.175
	4000	0.973	6.325	1.975
	125	0.987	0.005	0.053
	250	0.978	0.009	0.069
STI	500	0.931	0.018	0.098
	1000	0.908	0.019	0.098
	2000	0.962	0.020	0.098
	4000	0.956	0.017	0.089
	General	0.964	0.000	0.010
	Men	0.956	0.000	0.011
	Women	0.960	0.000	0.011

Shap values associated with certain architectural features indicate their stronger influence on acoustic performance. For instance, the impact of wall material on metrics such as C_{80} and D_{50} is particularly pronounced, suggesting that the choice of wall material significantly affects sound diffusion and clarity in the space. Furthermore, the observed influence of room length in the X direction on metrics like C_{80} and EDT at medium frequencies underscores the importance of room dimensions in shaping acoustic characteristics. Receiver position also emerges as a key factor, especially at medium frequency, highlighting its role in sound distribution within the space. However, it's important to delve deeper into the underlying mechanisms driving these observations. Future research could explore the physical properties of materials, such as their absorption coefficients and reflection properties, to better understand their impact on acoustic metrics. Additionally, investigating the interaction

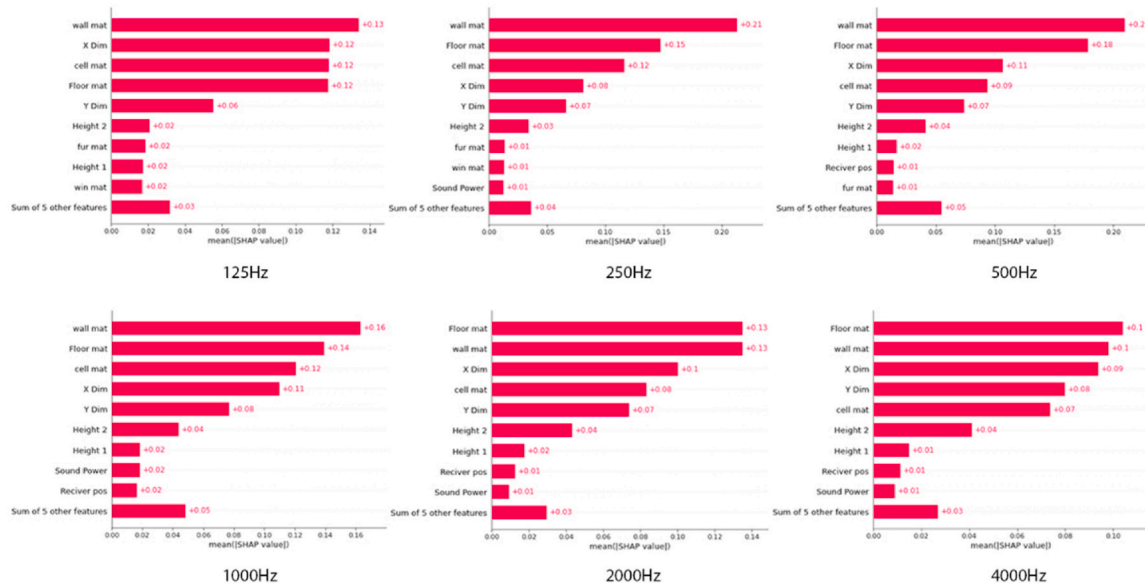
between architectural features and sound propagation dynamics could provide further insights into the observed effects.

Lastly, it should be noted as a restriction that the diffusion coefficient of all the materials in the current research has been considered as a single number for all frequencies to simplify the simulation process, it is recommended to add more variations in the diffusion coefficients corresponding to different frequencies for a more accurate and comprehensive simulation.

4. Conclusion

The complex factors affecting the acoustic quality inside buildings are challenging to accurately incorporate in the building design process, particularly in the early stages. As a result, analytical methods appear beneficial for enhancing acoustic quality in educational buildings, an aspect that recent studies have not thoroughly explored. Additionally, comparing various design options for indoor acoustics can optimize early building sketches more effectively, particularly in educational buildings where a range of acoustic activities occur. The present research investigates new methods for evaluating key architectural factors that impact indoor acoustic performance. Artificial intelligent algorithms have been developed to compute acoustic parameters like T_{30} , EDT, C_{80} , D_{50} , and STI across different architectural designs. Furthermore, a deep-learning approach using conditional Generative Adversarial Networks has been created to produce heat maps of STIs during the early design phases. This method's primary contribution is to substitute the time-consuming simulations typically employed in indoor acoustic analysis. The Pix2Pix model's generated 51 test results show SSIM values ranging from 15.56 to 31.57 and PSNR values between 0.81 and 0.97. Also, the Catboost model presents the R2 value of the T_{30} , EDT, C_{80} , D_{50} , and STI almost more than 0.9 % in different frequencies and modes. Alongside the predictive model, a feature importance analysis has been carried out to evaluate the impact of each architectural element on indoor acoustic indexes. The results indicate that the various architectural features impact the model's output to different extents.

Furthermore, the models introduced in this study serve as effective supplements to prescriptive guidelines by analyzing the real impact of architectural elements on the indoor acoustic indices of educational buildings. The dataset created for this research comprises 336 STI heatmaps and 980 numerical indoor acoustic plans generated through a parametric process using Pachyderm and Grasshopper. As ray tracing

**Fig. 6.** SHAP value of each recorded feature on the T_{30} .

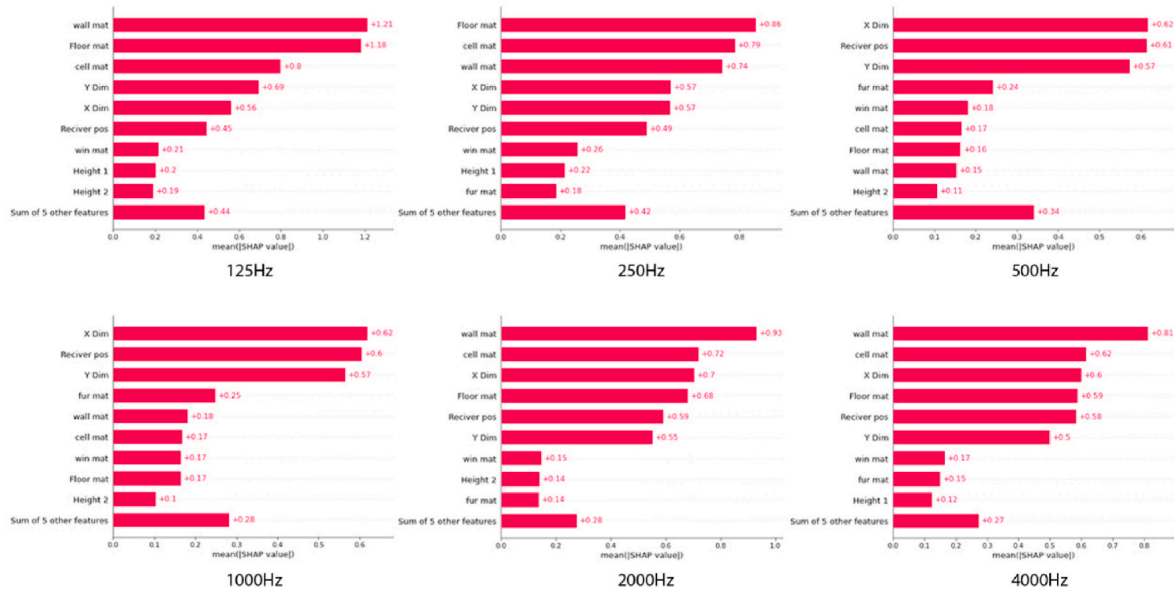


Fig. 7. SHAP value of each recorded feature on the C₈₀.

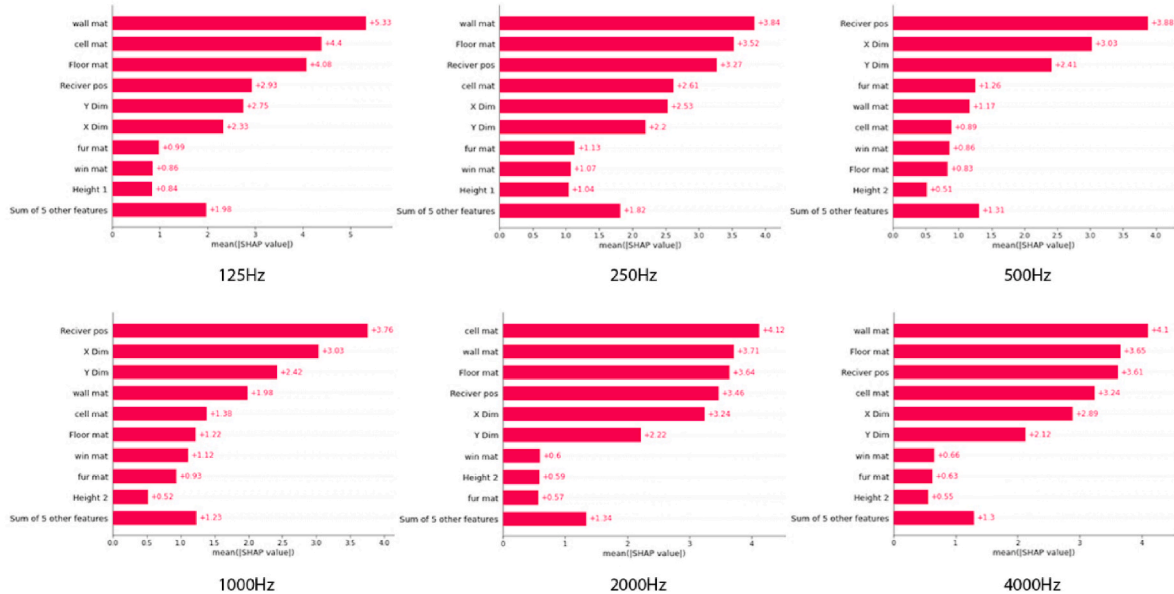


Fig. 8. SHAP value of each recorded feature on the D₅₀.

and image source simulation are time-intensive, the dataset generation is notably time-consuming. Currently, the applicability of these results is confined to a narrow range of architectural layouts characterized by right-angled and rectangular shapes, spanning sizes from 24 to 672 square meters. However, as the size and diversity of the generated architectural plans grow to include T-shaped, L-shaped, U-shaped, and Z-shaped designs, the results will be applicable to a broader array of scenarios. This research is poised to assist designers, their clients, and possibly Authority Having Jurisdictions (AHJs) during the preliminary design phases by offering a preview of various design alternatives for more in-depth analysis. Additionally, this tool could aid classroom teachers in assessing and modifying architectural factors like furniture placement or source location to enhance acoustic quality. Future research could also explore the impact of this tool on actual classroom settings and students' subjective speech intelligibility. In alignment with the architects' initial objectives of applying specific constraints during the design process, the methodology presented can also be adapted to

any dataset that architects wish to develop.

CRediT authorship contribution statement

Mohammad Tabatabaei Manesh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Conceptualization. **Arman Nikkhah Dehnavi:** Writing – review & editing, Validation, Methodology, Data curation. **Mohammad Tahsildoost:** Writing – review & editing, Supervision, Conceptualization. **Pantea Alambeigi:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

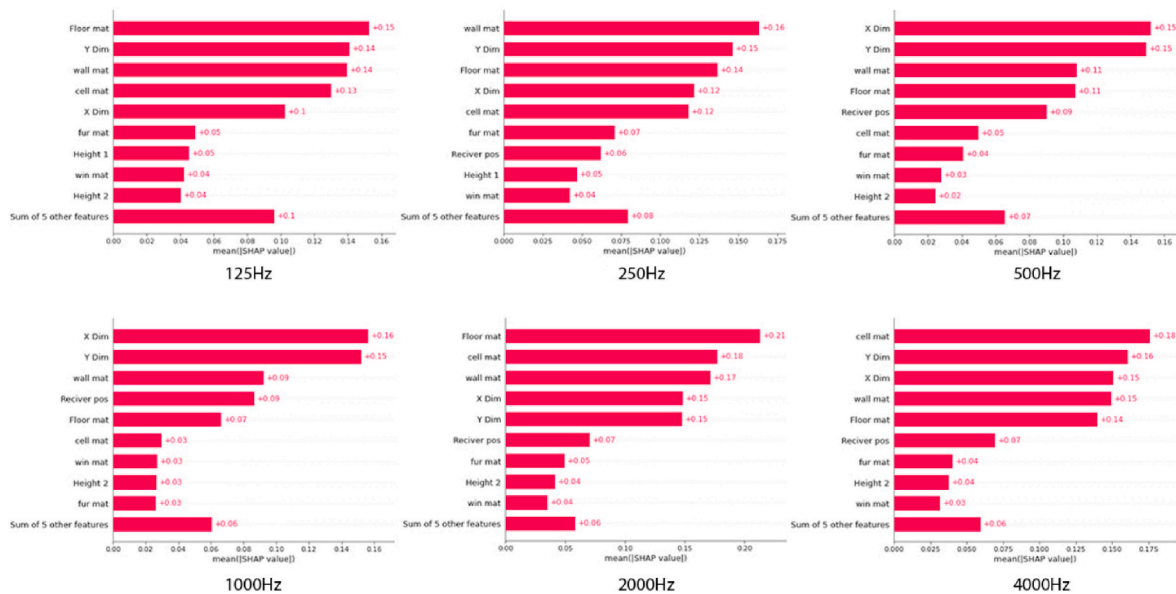


Fig. 9. SHAP value of each recorded feature on the EDT.

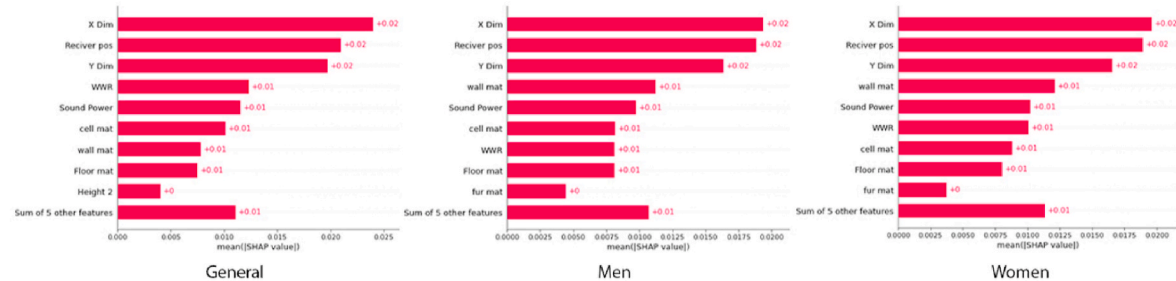


Fig. 10. SHAP value of each recorded feature on the STI.

Data availability

I have shared the link to my data at the manuscript.

Acknowledgements

This research did not receive external financial support, the collective efforts of all authors played a crucial role in bringing this project to fruition.

References

- [1] N.G. Vardaxis, D. Bard, K. Persson Wayne, Review of acoustic comfort evaluation in dwellings—part I: associations of acoustic field data to subjective responses from building surveys, *Build. Acoust.* 25 (2) (2018) 151–170, <https://doi.org/10.1177/1351010X18762687>.
- [2] J.T. Anderson, D. Van Holliday, R. Kloser, D.G. Reid, Y. Simard, Acoustic seabed classification: current practice and future directions, *ICES (Int. Coun. Explor. Sea) J. Mar. Sci.* 65 (6) (2008) 1004–1011, <https://doi.org/10.1093/icesjms/fsn061>.
- [3] P.M. Bluyssen, D. Zhang, S. Kurvers, M. Overtom, M. Ortiz-Sanchez, Self-reported health and comfort of school children in 54 classrooms of 21 Dutch school buildings, *Build. Environ.* 138 (March) (2018) 106–123, <https://doi.org/10.1016/j.buildenv.2018.04.032>.
- [4] W. Babisch, The noise/stress concept, risk assessment and research needs, *Noise Health* 4 (16) (2002) 1–11.
- [5] J. Kang, et al., Ten questions on the soundscapes of the built environment, *Build. Environ.* 108 (2016) 284–294, <https://doi.org/10.1016/j.buildenv.2016.08.011>.
- [6] A. Astolfi, et al., Influence of classroom acoustics on noise disturbance and well-being for first graders, *Front. Psychol.* 10 (2019) 2736, <https://doi.org/10.3389/fpsyg.2019.02736>.
- [7] B. Shield, J. Dockrell, The effects of environmental and classroom noise on the academic attainments of primary school children, *J. Acoust. Soc. Am.* 123 (2008) 133–144, <https://doi.org/10.1121/1.2812596>.
- [8] P. Alambeigi, J. Burry, S. Zhao, E. Cheng, A study of human vocal effort in response to the architectural auditory environment, *Architect. Sci. Rev.* 63 (3–4) (2020) 262–274.
- [9] M. Kob, G. Behler, A. Kamproff, O. Goldschmidt, C. Neuschaefer-Rube, Experimental investigations of the influence of room acoustics on the teacher's voice, *Acoust. Sci. Technol.* 29 (1) (2008) 86–94.
- [10] C.J. Nudelman, P. Bottalico, L.C. Cantor-Cutiva, The Effects of Room Acoustics on Self-Reported Vocal Fatigue: a Systematic Review, *Journal of Voice*, 2023.
- [11] D. Zhang, M.A. Ortiz, P.M. Bluyssen, Clustering of Dutch school children based on their preferences and needs of the IEQ in classrooms, *Build. Environ.* 147 (2019) 258–266, <https://doi.org/10.1016/j.buildenv.2018.10.014>.
- [12] A. Hamida, D. Zhang, M.A. Ortiz, P.M. Bluyssen, Indicators and methods for assessing acoustical preferences and needs of students in educational buildings: a review, *Appl. Acoust.* 202 (2023) 109187, <https://doi.org/10.1016/j.apacoust.2022.109187>.
- [13] M. Choi, M. Kapur, Human-centered design in acoustics education for undergraduate music majors, *J. Acoust. Soc. Am.* 151 (4) (Apr. 2022) 2282–2289, <https://doi.org/10.1121/10.0010043>.
- [14] J.S. Bradley, H. Sato, The intelligibility of speech in elementary school classrooms, *J. Acoust. Soc. Am.* 123 (4) (Apr. 2008) 2078–2086, <https://doi.org/10.1121/1.2839285>.
- [15] P. Croce, F. Leccese, G. Salvadori, U. Berardi, Proposal of a simplified tool for early acoustics design stage of classrooms in compliance with speech intelligibility thresholds, *Energies* 16 (2023) 813, <https://doi.org/10.3390/EN16020813>, vol. 16, no. 2, p. 813, Jan. 2023.
- [16] Association of Australasian Acoustical Consultants, Association of Australasian Acoustical Consultants Guideline for Educational Facilities Acoustics, 2010, pp. 1–15.
- [17] K. Ueno, T. Fukuchi, Y. Tsuchiya, Revision of Standard for Sound Environment in School Buildings Published by the Architectural Institute of Japan, *eForum Acusticum*, 2020, pp. 177–180.
- [18] Richard Daniels, Acoustic design of schools: performance standards, February, *Building Bulletin* 93 (93) (2015) 43 [Online]. Available: <https://www.gov.uk/government/publications/bb93-acoustic-design-of-schools-performance-standards>.

- [19] American National Standards Institute, ANSI/ASA S12.60-2010/Part 1. *Acoustical Performance Criteria, Design Requirements, and Guidelines for Schools, Part 1: Permanent Schools*, vol. 44, Ansi/Asa, 2010.
- [20] W. Mikulski, J. Radosz, Acoustics of classrooms in primary Schools - results of the reverberation time and the speech transmission index assessments in selected buildings, *Arch. Acoust. Q.* 36 (4) (2011) 777–793, <https://doi.org/10.2478/V10168-011-0052-6>.
- [21] L.L. Beranek, T.J. Mellow, Chapter 10 - sound in enclosures, in: L.L. Beranek, T. J. Mellow (Eds.), *Acoustics: Sound Fields and Transducers*, Academic Press, 2012, pp. 449–479, <https://doi.org/10.1016/B978-0-12-391421-7.00010-5>.
- [22] N.G. Vardaxis, D. Bard, Review of acoustic comfort evaluation in dwellings: Part III—airborne sound data associated with subjective responses in laboratory tests, *Build. Acoust.* 25 (4) (2018) 289–305, <https://doi.org/10.1177/1351010X18788685>.
- [23] L.L. Beranek, *Acoustic Measurements*, second ed., American Institute of Physics, 1988.
- [24] A.G. Prinn, *A Review of Finite Element Methods for Room Acoustics*, 2023, pp. 367–395.
- [25] S. Kirkup, *The Boundary Element Method in Acoustics*, vol. 8, 2007.
- [26] L. Savioja, U.P. Svensson, Overview of geometrical room acoustic modeling techniques, *J. Acoust. Soc. Am.* 138 (2) (2015) 708–730.
- [27] R.A. Rathnayake, W.K.I.L. Wanniarachchi, Image source method based acoustic simulation for 3-D room environment, *International Journal of Scientific and Technology Research* 8 (11) (2019) 222–228.
- [28] M.J. Bianco, et al., Machine learning in acoustics: theory and applications, *J. Acoust. Soc. Am.* 146 (5) (2019) 3590–3628, <https://doi.org/10.1121/1.5133944>.
- [29] J. Nannariello, F. Fricke, The prediction of reverberation time using neural network analysis, *Appl. Acoust.* 58 (3) (1999) 305–325, [https://doi.org/10.1016/S0003-682X\(98\)00081-4](https://doi.org/10.1016/S0003-682X(98)00081-4).
- [30] R.F. Pérez, *Machine-learning-based Estimation of Room Acoustic Parameters*, Aalto University Master's Thesis, 2018.
- [31] R. Abarghoie, Z.S. Zomorodian, M. Tahsildoost, Z. Shaghaghian, A machine-learning framework for acoustic design assessment in early design stages, *CoRR abs/2109.06459* (2021) [Online]. Available: <https://arxiv.org/abs/2109.06459>.
- [32] C.Y. Yeh, Y.S. Tsay, Using machine learning to predict indoor acoustic indicators of multi-functional activity centers, *Appl. Sci.* 11 (12) (2021), <https://doi.org/10.3390/app11125641>.
- [33] Manesh Tabatabaei, Mohammad, Nikkhah Dehnavi, Arman, Mohammad Tahsildoost, Pantea Alambeigi, Machine Learning-Based Prediction of Acoustic Indicators for Enhanced Educational Building Comfort, *Mendeley Data*, V1, 2024, <https://doi.org/10.17632/yr3pr7bz8.1>.
- [34] Andrew Marsh, *Ecotect, Version 5.20(b)*, 2003.
- [35] ODEON A/S, *ODEON Room Acoustics Software: User's Manual*, Version 18, pp. 81–83, November 23, 2023. Available: <https://odeon.dk/downloads/user-manual/>.
- [36] J. Holger Rindel Claus Lynge Christensen, A. Christian Gade Gade, M. Akustik, and H. Edvard Teglers Vej, "Dynamic Sound Source for Simulating the Lombard Effect in Room Acoustic Modeling Software".
- [37] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: from error visibility to structural similarity, *IEEE Trans. Image Process.* 13 (4) (2004) 600–612, <https://doi.org/10.1109/TIP.2003.819861>.
- [38] A. Horé, D. Ziou, Image quality metrics: PSNR vs. SSIM, in: 2010 20th International Conference on Pattern Recognition, 2010, pp. 2366–2369, <https://doi.org/10.1109/ICPR.2010.579>.
- [39] L. Prokhorenkova, G. Gusev, A. Vorobev, A.V. Dorogush, A. Gulin, Catboost: unbiased boosting with categorical features, *Adv. Neural Inf. Process. Syst.* 2018-Decem (4) (2018) 6638–6648.
- [40] M. Steurer, R.J. Hill, N. Pfeifer, Metrics for evaluating the performance of machine learning based automated valuation models, *J. Property Res.* 38 (2) (Apr. 2021) 99–129, <https://doi.org/10.1080/09599916.2020.1858937>.
- [41] S.M. Lundberg, S.I. Lee, A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems*, 2017, pp. 4766–4775 [Online]. Available: <https://github.com/slundberg/shap>. (Accessed 24 January 2023).
- [42] D. Jia, L. Yang, X. Gao, K. Li, Assessment of a new solar radiation nowcasting method based on FY-4A satellite imagery, the McClear model and SHapley additive exPlanations (SHAP), *Rem. Sens.* 15 (2023) 2245, <https://doi.org/10.3390/RS15092245>, vol. 15, no. 9, p. 2245, Apr. 2023.
- [43] U. Sara, M. Akter, M.S. Uddin, Image quality assessment through FSIM, SSIM, MSE and PSNR—a comparative study, *J. Comput. Commun.* 7 (3) (2019) 8–18, <https://doi.org/10.4236/jcc.2019.73002>.